

The empirical recurrence for OEIS A235219: recovering a conjecture that is not written down

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Abstract

OEIS A235219 records “Empirical recurrence of order 59”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 69$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 59, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 59 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A235219 is “Number of $(n+1) \times (4+1)$ 0..2 arrays with 2×2 edge jumps all no more than +1 in one of the clockwise or counterclockwise directions or both.”. It has offset 1 and begins

21693, 2667851, 306034401, 35761397619, 4160135051925, 484500703182059, 56410299169343241, 6568304367121

The entry states:

Empirical recurrence of order 59 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 08:29:36, revision 7) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 244$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 69$ of the 244, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 69$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S = 488$ exact terms of the model returns order 59 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{59} c_i a(n-i),$$

c_1	101	c_{16}	19809828791265993	c_{31}	771781673132485445874	c_{46}	14434188900053
c_2	2648	c_{17}	11239392372036384	c_{32}	452083657358640794737	c_{47}	4789369464878
c_3	−94517	c_{18}	−233493492791198288	c_{33}	−983780791094861507434	c_{48}	−1938515081916
c_4	−798034	c_{19}	6213609417841041	c_{34}	−285839852729365768007	c_{49}	−342747969943
c_5	32030316	c_{20}	1935986425545690509	c_{35}	904249854373059582360	c_{50}	1695341272768
c_6	−29297191	c_{21}	−844667477146074504	c_{36}	80935218286884781519	c_{51}	127903647999
c_7	−3639799974	c_{22}	−11403631098195414399	c_{37}	−601119171684471676990	c_{52}	−92263846017
c_8	13385904660	c_{23}	8431842093063474698	c_{38}	31040510396199787312	c_{53}	−6306482981
c_9	206937233196	c_{24}	47884127142218585015	c_{39}	288756143080872544209	c_{54}	28687488066
c_{10}	−1081176919517	c_{25}	−47060317108073335818	c_{40}	−44367600322335752307	c_{55}	−1134235037
c_{11}	−6843622934498	c_{26}	−143180839791286405940	c_{41}	−99699354933926299227	c_{56}	−4319451943
c_{12}	45051376881814	c_{27}	171985820700807610536	c_{42}	22945719816112017650	c_{57}	2995073045
c_{13}	139418662565999	c_{28}	303242240726362927672	c_{43}	24463666979771797613	c_{58}	209979932
c_{14}	−1159000771173620	c_{29}	−433448251482283400645	c_{44}	−7121074233645553370	c_{59}	−17768139
c_{15}	−1732921522506836	c_{30}	−449204589931672857695	c_{45}	−4178932412819547960		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 59, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $59 + 4$ terms removed returns order 59 as well, so $\deg q_{\min} = 59 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $59 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 59 that this sequence satisfies — and that family turns out to have exactly one member.

References

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